MODUPE AWOJUYIGBE

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EDUCATION

The University of Texas at San Antonio *Expected: May 2027*

- Bachelor of Science in **Computer Engineering**– GPA: **3.7/4.0**

Achievements: NSBE Scholar, Klesse Scholar, USAA Scholar, Society of Women Engineers, herScript, Rewriting the Code
Coursework: Computer Architecture, C++ and Data Structures, Engineering Analysis II, Microcomputer Systems

SKILLS

- **Programming Languages:** Python, C, C++, JavaScript, HTML, CSS
- **Embedded Systems & Hardware:** Microcontrollers, Arduino, Sensor Integration, Servo Motor Control, Circuit Analysis, 3D Printing
- **Software, Data and, Validation Tools:** Pygame, Matplotlib, Pandas, Linux, Ubuntu, SCADA Systems, Test Plan Review, Machine Data Validation, Software Flashing, Product Status Reports
- **Professional Strengths:** Cross-Functional Collaboration, Adaptability, Problem Solving, Attention to Detail

EXPERIENCE

Caterpillar Inc.
Autonomy Engineering Intern, Peoria, IL **May 2026 – Present**

- Built a multi machine autonomy simulation using **Python**, Pygame, and Matplotlib to model construction machine behavior, test real world scenarios, evaluate operational tradeoffs for safer autonomous operation.
- Supported validation of autonomous machine features by reviewing test plans, validating machine data, performing safety checks, and improving customer demo readiness by **25%**.
- Flashed software, ran product status reports, and operated machines through a remote-control operator assist station while troubleshooting electrical issues that improved machine performance by **20%**.

EDP Renewables
Operations Engineering Intern, Houston, TX **June 2025 – August 2025**

- Designed a standardized data template that reduced project setup delays by **70%**, streamlining cross-functional coordination between contractors and engineering teams.
- Engineered a **Python** automation tool (Selenium) to aggregate weather data, reducing reporting time by **50%**.
- Reduced company spend by **$20,000** in operational costs via automated tools and data logging solutions that improved workflow and visibility.

The University of Texas at San Antonio
Teaching Assistant / Peer Educator, San Antonio, TX **August 2024 – Present**

- Mentored **50+** students in STEM related courses, translating complex engineering concepts into real-world applications, which improved student pass rates by **60%**.

TECHNICAL PROJECTS

Robotic Arm Project **Spring 2026**

- Programmed an Arduino based robotic arm in **C++,** integrating ultrasonic sensors and servo motors to control object detection and improve pickup accuracy by **60%**.

Real-Time Voice Captioning and Translation System **Spring 2025**

- Built a real-time captioning and translation system using a **PIC microcontroller** and **Python** to convert speech into formatted text on a Liquid Crystal Display.
- Developed a Python speech-to-text pipeline using **Cloud Translation API** and **Gemini** for contextual understanding and multilingual translation across 100+ languages.

Autonomous Line Tracking Vehicle **Fall 2024**

- Engineered an autonomous mini vehicle utilizing infrared sensors and **C++** programming to enable precise path detection and following capabilities.
- Designed and implemented a control system architecture integrating microcontrollers with motor drivers to achieve responsive navigation with **<2ms** latency.

LEADERSHIP

National Society of Black Engineers (NSBE) – *Vice President* **May 2026 – Present**

- Led executive board operations for **14** officers by managing agendas, accountability tracking, and chapter goal updates, improving task follow through by **25%** and increasing active chapter participation.

The University of Texas at San Antonio – *Resident Assistant* **May 2024 – August 2024**

- Supervised a community of **150+** residents while planning orientation events, improving safety compliance across 5 residence halls, and increasing first year student satisfaction by **30%.**